Fiscal Year:	FY 2020	**Task Last Updated:**	FY 07/14/2023
PI Name:	Hong, Pengyu Ph.D.		
Project Title:	Predicting Individual Differences in Learning and Performance Using Machine Learning Methods for Dynamic Stabilization in a Spaceflight Analog Task		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hongpeng@brandeis.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	781-736-2729
Organization Name:	Brandeis University		
PI Address 1:	Brandeis University Volen Center for Complex Systems 135		
PI Address 2:	MS 018, Brandeis University		
PI Web Page:			
City:	Waltham	**State:**	MA
Zip Code:	02453	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Synergy
Start Date:	10/01/2018	**End Date:**	09/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dizio, Paul Ph.D. (Brandeis University)		
Grant/Contract No.:	NNX16AO69A-SYN0004		
Performance Goal No.:			
Performance Goal Text:			
	Synergy Project REPORT SUBMITTED DECEMBER 2019 This is a synergy project with Translational Research Institute for Space Health (TRISH) postdoctoral fellow Dr. Vivekanand Pandey Vimal. To study the feasibility of our proposal, we trained a multilayered perceptron on previously collected data of 30 subjects in our spaceflight analog task. This preliminary approach achieved 80% accuracy in a leave-one-out test, where data from mid-learning on the first day of experimentation was used to predict performance at the end of the second day of experimentation. Therefore we are confident that 40 more subjects should provide better and more robust results. In our current plan we will begin by using random forest and boosting tree methods to		

Task Description:	determine which of our metrics are redundant and which are the critical metrics for quantifying learning and performance. Next, we will use recurrent neural networks to build a more robust classifier that can predict final performance using data from earlier in the experiment. Because our project is exploratory, we will add more subjects if it is needed and try different approaches if these are not ideal. Our paradigm requires subjects to learn to dynamically stabilize themselves in a spaceflight analog condition where they cannot rely on relevant gravitational cues to determine their angular position. We have found significant individual differences where some subjects show robust learning and good performance whereas others actually worsen with exposure and report spatial disorientation. Our assessment takes two experimental sessions across consecutive days and requires the use of our Multi-Axis Rotation System. The machine learning approach will allow us to distinguish which participants will become worse with time and which ones will show significant learning. Once we have these classifiers, after only a short exposure, we will be able to predict which trainees will be highly effective at maintaining orientation and which ones will be susceptible to spatial disorientation. Those that will be identified as being susceptible will be given additional training. Recent results from our Specific Aim 2 reveal our novel training program can lead to enhanced performance in subjects that are prone to spatial disorientation, even those that traditionally would become worse with exposure. These results also revealed that every participant retained their abilities even after 4 months, suggesting that the classifiers could predict performance of the astronauts as they reach Mars, based on tests in a spaceflight analog task on Earth. As a future direction, we aim to use these classifiers to identify which variables weigh heavily in successful trainees and design training and cuing that specifically enhances these variables in all trainees. INITIAL INFORMATION WHEN PROJECT INITIATED IN OCTOBER 2018 This project aims to develop and customize the machine learning architecture for a Space Flight Analog Environment at Brandeis University (Paul Dizio, Vivekanand Vimal). Aims: To determine which mathematical metrics are the core metrics that predict performance and learning. Also, determine which metrics are redundant and dependent on other metrics. To determine whether the same classifier can be generalized to different conditions and paradigms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As described in the previous section, impacts and benefits: 1. A machine learning technique for clustering individuals in performance of a complex skilled motor learning paradigm without the need for human labeling. 2. The ability to predict with 80% accuracy a person's final ability to dynamically stabilize in a spaceflight analog task using their performance in early training stage. 3. The ability to identify sub-optimal strategies that can be reduced by targeted training. Our research results can be applied to develop personalized training of astronauts. Imaging an application scenario that we need to train astronauts, who are already in deep space, to perform certain new tasks. To maximize the training outcomes and minimize training risks, we can collect data from astronauts in simulation environments on Earth, use those data to develop predication models, transmit the models to the space crafts to better guide the training of those astronauts in deep space.
Task Progress:	REPORT SUBMITTED DECEMBER 2019 Our overall objective was to characterize, identify, and predict individual differences in a skilled motor learning task in a spaceflight analog condition. We created a spaceflight analog condition by strapping 37 blindfolded subjects into a device that behaved like an inverted pendulum in the horizontal roll plane, where subjects could only rely on motion information. Subjects were instructed to use a joystick to stabilize themselves across two experimental sessions on consecutive days. Similar to astronauts, they could not use gravitational cues to determine their angular position and many soon became spatially disoriented. In the spaceflight analog task, most demonstrated minimal learning, poor performance, and a characteristic pattern of positional drifting during horizontal roll plane balancing. To understand the wide range of individual differences observed, we used a Bayesian Gaussian Mixture method to cluster subjects into 3 statistically distinct groups that represent Proficient, Somewhat Proficient, and Not Proficient performance. We found that subjects in the Not Proficient group exhibited a suboptimal strategy of using very stereotyped large magnitude joystick deflections. We also used a Gaussian Naive Bayes method to create predictive classifiers. As early as the second block of experimentation (out of 10), we could predict a subject's final group with 80% accuracy. Our findings indicate that machine learning can help predict individual performance and learning in a disorienting dynamic stabilization task and identify suboptimal strategies in not proficient subjects, which could lead to personalized and more effective training programs.
Bibliography Type:	Description: (Last Updated: 07/14/2023)
Articles in Peer-reviewed Journals	Vimal VP, Zheng H, Hong P, Fakharzadeh LN, Lackner JR, DiZio P. "Characterizing individual differences in a dynamic stabilization task using machine learning." Aerosp Med Hum Perform. 2020 Jun 1;91(6):479-88. https://doi.org/10.3357/AMHP.5552.2020 ; PMID: 32408931 , Jun-2020